Neural Network Field Guide

42 architectures and training schemes, from the perceptron (1958) to diffusion LLMs (2025).

Each diagram draws a network in miniature, from input at the bottom to output at the top. In most panels, a circle is one neuron; in sequence models, where each column is a token, it is that token's whole vector at one layer. A square is larger: an image patch, a matrix-valued state, a whole network; panels showing a way of training draw networks as gray boxes. A cell's color and mark both show its role, so the diagrams read without color. Faded parts play little or no role in the step shown. Parts repeated across positions, windows, copies, or passes share weights; layers, experts, encoders, and decoders drawn alike do not. Years and training methods are from landmark papers, not always the first; many can be trained other ways.

Input
- Input unit
- Image patch
- Noisy input
- Random noise
- Visible unit (in and out)
- Masked token

Hidden
- Hidden unit
- Probabilistic hidden
- Capsule (vector output)
- Router

State
- Recurrent unit
- Memory cell
- Matrix-valued state

Mixing
- Attention
- Convolution or pooling

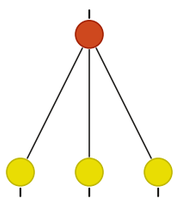
Output
- Output unit
- Reconstruction of input
- Velocity

Network
- A whole network

Links
- Connection
- Skip connection
- Data passed on or fed back
- Gate, acting at the dot
- Cross-attention, comparison
- Learnable function

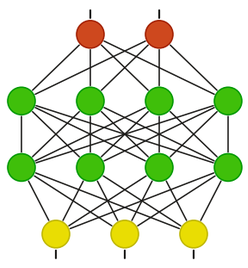
FEED-FORWARD AND CONVOLUTIONAL

Perceptron



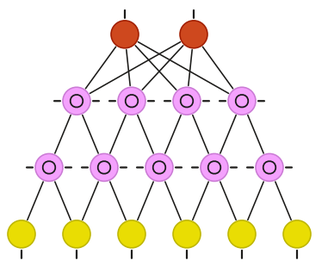
Weighted sum, then a threshold
1958 · supervised

MLP



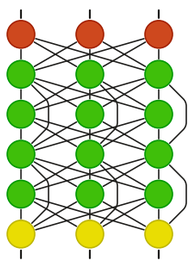
Stacked dense layers
1986 · supervised

CNN



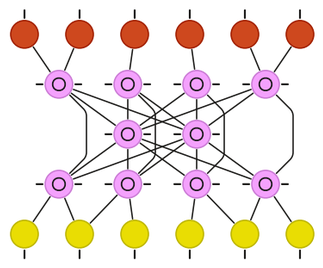
Local filters, shared weights
1989 · supervised

ResNet



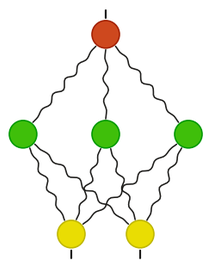
Blocks add to a skip path
2015 · supervised

U-Net



Shrink, expand, skip across
2015 · supervised

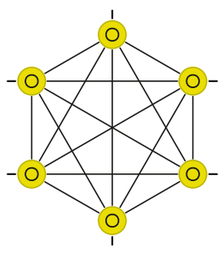
KAN



Learnable functions on edges
2024 · supervised

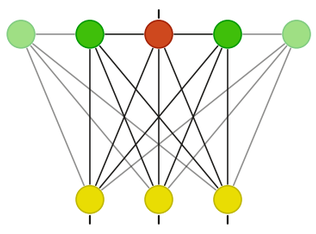
ENERGY-BASED AND SELF-ORGANIZING

Hopfield network



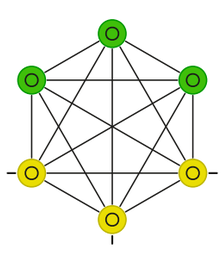
Settles toward a stored pattern
1982 · hebbian

Kohonen map



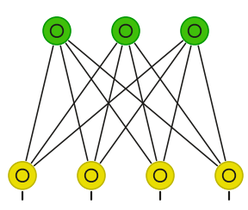
Winner and neighbors move
1982 · competitive

Boltzmann machine



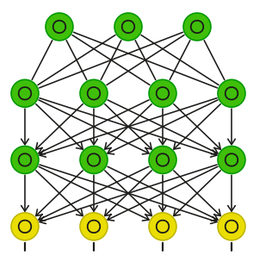
Stochastic, fully connected
1985 · likelihood

RBM



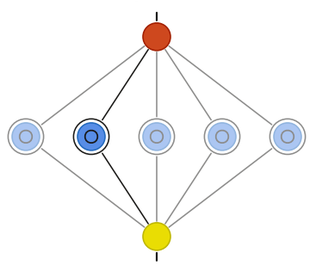
No links within a layer
1986 · likelihood

Deep belief network



Stacked RBMs, trained greedily
2006 · likelihood

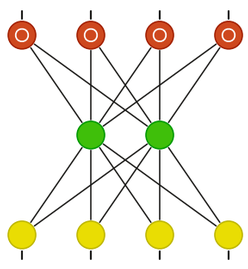
Modern Hopfield network



Attention as memory lookup
2020 · supervised

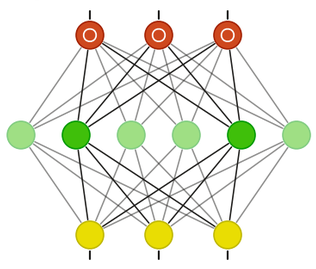
AUTOENCODERS AND GENERATIVE

Autoencoder



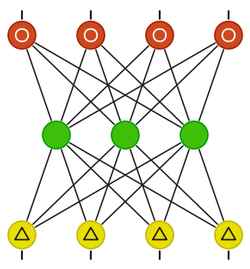
Squeeze, then reconstruct
1986 · reconstruct

Sparse autoencoder



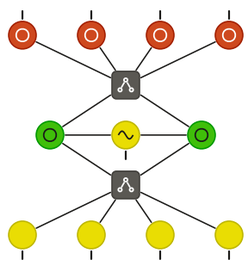
Wide code, few active units
2006 · reconstruct

Denoising autoencoder



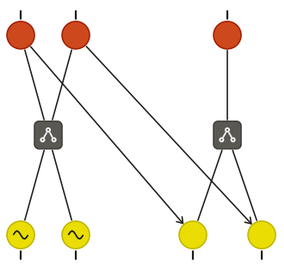
Clean up a noisy input
2008 · denoise

Variational autoencoder



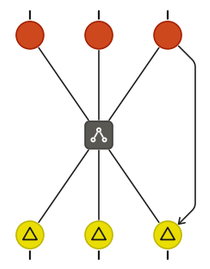
Sample a code, then decode
2013 · variational

GAN



Generator vs. discriminator
2014 · adversarial

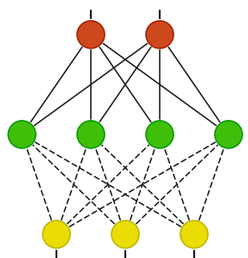
Diffusion model



One denoiser, run T times
2015 · variational

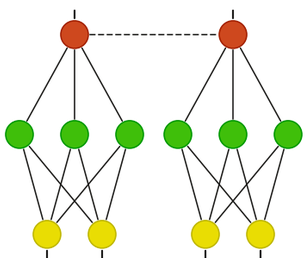
SIMILARITY

Radial basis network



Distance to stored centers
1988 · supervised

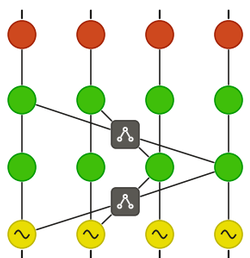
Siamese network



Twin towers, shared weights
1993 · contrastive

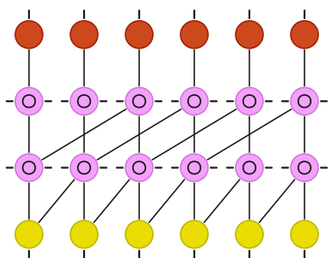
FLOWS AND AUTOREGRESSIVE MODELS

Normalizing flow



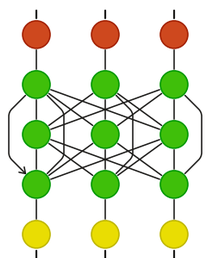
Invertible map from noise to data
2014 · likelihood

WaveNet



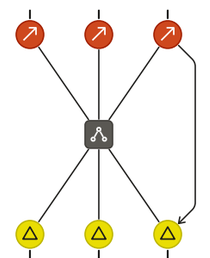
Dilated causal convolutions
2016 · next token

Neural ODE



A flow with continuous depth
2018 · likelihood

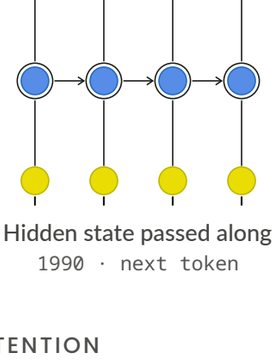
Flow matching



Learns a velocity, noise to data
2022 · flow matching

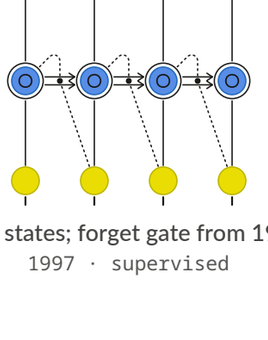
RECURRENT AND MEMORY

RNN



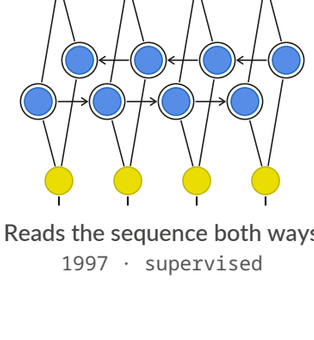
Hidden state passed along
1990 · next token

LSTM



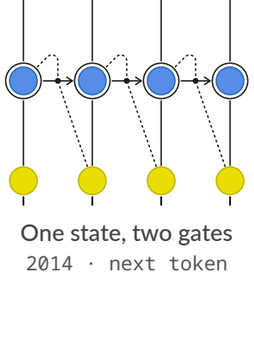
Two states; forget gate from 1999
1997 · supervised

Bidirectional RNN



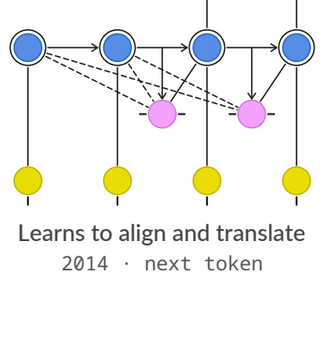
Reads the sequence both ways
1997 · supervised

GRU



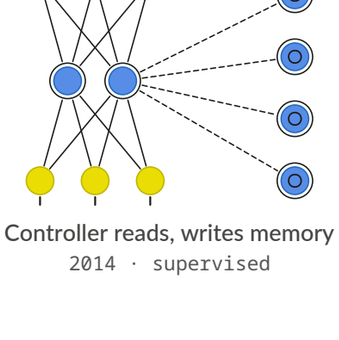
One state, two gates
2014 · next token

Seq2seq with attention



Learns to align and translate
2014 · next token

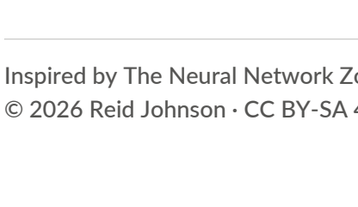
Neural Turing machine



Controller reads, writes memory
2014 · supervised

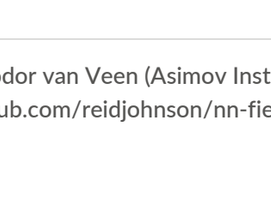
ATTENTION

Transformer



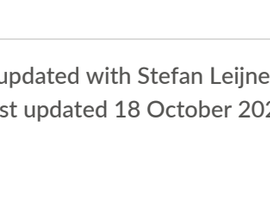
Encoder, decoder, cross-attention
2017 · next token

BERT



Every token sees every token
2018 · masked token

GPT-style LLM



Sees itself and earlier tokens
2018 · next token

ViT

Image patches become tokens
2020 · supervised

CLIP

Match images to captions
2021 · contrastive

Masked diffusion LLM

Unmask in parallel, step by step
2025 · masked token

LINEAR-TIME MODELS AND HYBRIDS

Linear attention

Softmax-free attention as an RNN
2020 · next token

Mamba

Selective state, linear time
2023 · next token

Mamba–attention hybrid

Mostly recurrent, some attention
2024 · next token

ROUTING AND GRAPHS

Graph neural network

Nodes pool neighbor messages
2005 · supervised

Sparse MoE

Each token → its top-k experts
2017 · next token

Capsule network

Vectors routed by agreement
2017 · supervised

Inspired by The Neural Network Zoo by Fjodor van Veen (Asimov Institute, 2016; updated with Stefan Leijnen, 2019). www.asimovinstitute.org/neural-network-zoo

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